Channel Coding Techniques in Macroscopic Air-Based Molecular Communication

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ABSTRACT

Long distance molecular communication (MC) is prone to channel errors, leading to an increased bit error rate. For solving the issue, a new coding scheme named weak sequence preventive mapping code is designed, conforming to the channel characteristics of the macroscopic air-based MC using fluorescent dye. The scheme not only minimizes the transmission errors, but also conserves the resource in the form of sprayed water-based dye solution.

CCS CONCEPTS

• **Hardware → Wireless devices**.

KEYWORDS

Molecular communication, channel coding, testbed, fluorescence

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1 INTRODUCTION

Molecular communication (MC) is inspired by nature, where information is carried by molecules. MC is an emerging technological field that can be applied to many multiscale applications, where the performance of conventional communication techniques suffer. These include targeted drug delivery, lab-on-a-chip, monitoring of oil/gas pipelines and sewage systems, etc. [6].

Potential industrial applications of MC are investigated through testbed designs [2, 5]. Long distance transmission in MC, like any other scheme, is prone to errors leading to an increase in the bit error rate (BER). In order to tackle the issue, channel codes are implemented for effective and robust error detection and correction.

Channel codes have been studied for diffusion-based MC systems such as the inter-symbol interference (ISI)-free codes, where the BER is improved by mitigating the ISI [8]. A similar approach termed as the (modified) crossover resistant coding with time gap takes different types of molecules into account to further improve the BER performance [1]. Their performance, however, is limited in a three-dimensional diffusion-based MC system, where molecules could get lost. In [7] however, the focus is shifted to implementing channel codes in macro-scale MC testbeds.

This presented work furthers the effort to reduce the BER by introducing a new coding scheme named weak sequence preventive mapping (WSPM) code for air-based macroscopic MC testbed using water-based solution of fluorescent dye [2, 4]. Additionally, the resource requirement of WSPM code is less as compared to the other popular coding schemes from digital communications theory.

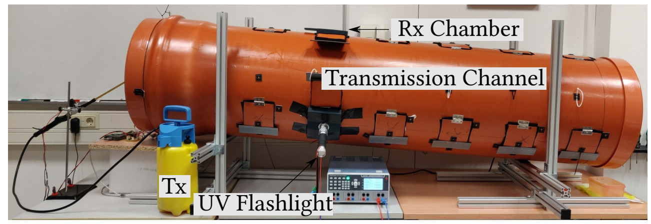

Figure 1: Macroscopic air-based MC testbed.

Table 1: Code table for (3,2) WSPM code.

$\boldsymbol{u}$	[0 0]	[0 1]	[1 0]	[1 1]
$\boldsymbol{x}$	[0 0 0]	[0 0 1]	[0 1 0]	[1 0 0]

2 TESTBED OVERVIEW

Figure 1 shows the macroscopic air-based MC testbed comprising of a 3 L transmitter (Tx) sprayer, injecting fluorescent uranine water-based solution into an industrial pipe (channel), at a pressure of 3 bar and a spray duration of $T_s = 20$ ms. At the receiver (Rx) side, the sprayed droplets are then detected by a high-speed digital camera. Under the influence of ultra-violet light, these droplets produce a peak emission at 548 nm, as described at length in [2, 4].

After applying the channel codes, the on-off keying (OOK) modulated info words (groups of info bits) are transformed into code words (groups of code bits), and sprayed from Tx with a bit duration of $T = 50$ ms. At the Rx, the recorded green pixel intensities are summed up and subjected to fixed threshold detection [4]. The detected code bits of the code words are then re-transformed back to obtain the original info words.

3 WEAK SEQUENCE PREVENTIVE MAPPING

A channel code adds redundant bits to a data sequence. An (n, k) coding scheme containing $n - k$ redundant bits is defined by its

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Table 2: Channel coding schemes comparison.

Channel Code	$d_{\min}$	R	Sprays per info bit
(2,1) RC	2	1/2	1
(3,2) SPC	2	2/3	0.75
(3,2) WSPM	1	2/3	0.375
(15,11) HC	3	11/15	0.682

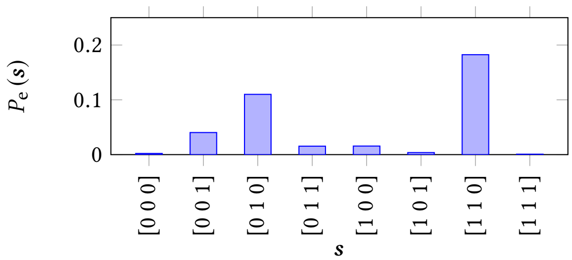

Figure 2: Sequence error analysis for $n = 3$ and $d = 178$ cm.

code rate R, which is the ratio of its info word length k to its code word length n. The error detection and correction capabilities of a channel code are determined by the minimum Hamming distance $d_{\min}$, i.e., the minimum number of positions at which two code words vary. To this end, a new channel coding scheme dubbed WSPM code is implemented for the first time in a macroscopic MC testbed, initially coined and designed for a diffusion-based microscopic MC system [3]. Here, a codebook for (n, k) block code is created by performing error analysis on all 2^n possible received words, out of which, 2^k least error-prone code words are selected and assigned to 2^k info words. Similarly, a (3,2) WSPM code is designed by analyzing a sequence length $n = 3$ over 50 trials, at distance $d = 178$ cm for a transmission sequence length of 100 OOK modulated bits, as shown in Figure 2. The selected code words in Table 1 reduce the probability of occurrence of the most error-prone sequence [1 1 0], and also make the coding scheme more resource efficient in terms of the required sprays per info bit on average.

4 RESULTS

Table 2 lists out a comparison of the minimum Hamming distance $d_{\min}$, the code rate R, and the average sprays per info bit of WSPM with respect to the implemented block codes from digital communications theory such as (2,1) repetition code (RC), (3,2) single parity check (SPC) code, and (15,11) Hamming code (HC). It is seen that (3,2) WSPM code uses the least amount of resource as compared to the other coding schemes, requiring only 0.375 sprays per info bit on average. This is even better than the uncoded OOK modulated scheme requiring on average 0.5 sprays per info bit. Additionally, the BER comparison of (3,2) WSPM code with the aforementioned block codes are carried out through testbed measurements of the code bits corresponding to 100 randomly generated OOK info bits at transmission distances $d = \{58, 88, 118, 148, 178\}$ cm over 20 trials. It is seen that (3,2) WSPM code performs the best with respect to BER as compared to the other implemented channel codes with higher $d_{\min}$ and/or lower R. This is because when two possible (3,2)

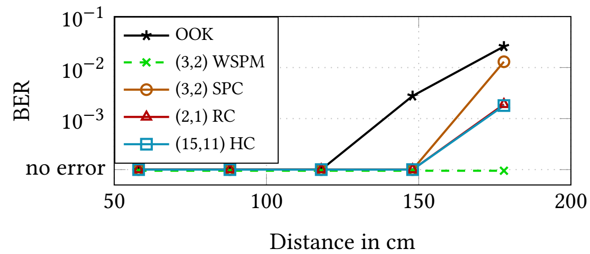

Figure 3: BER comparison of (3,2) WSPM code with block codes and uncoded OOK scheme.

WSPM code words are transmitted one after another, the likelihood of occurrence of the most error-prone sequence [1 1 0] in the stream of transmitted bits is merely 6.25%. This is for example, significantly less as compared to (3,2) SPC code, where the likelihood of occurrence of the same sequence is 75%.

5 CONCLUSION

In this work, (3,2) WSPM coding scheme is implemented for the first time in an air-based macroscopic MC testbed using fluorescent dye. It is seen that (3,2) WSPM code not only helps improve the BER performance as compared to the other implemented block codes from the literature, but also requires less resource in the form of sprayed water-based dye solution on average.

Future work will include incorporation of relays with channel codes for increasing the transmission lengths at low BER.

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